



RNA-seq DESeq2 Differential Expression and QC Workflow

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Overview

This workflow demonstrates a reproducible RNA-seq differential expression and reporting pipeline using the public GEO dataset GSE203539.

Raw sequencing data were originally processed through a Galaxy-based RNA-seq pipeline including quality assessment, read trimming, alignment to the GRCh39 reference genome, and feature-level quantification prior to downstream analysis.

This repository focuses on downstream quality control, outlier sensitivity assessment, DESeq2 differential expression analysis, and structured result export workflows in R.

Analysis Plan

1. Load RNA-seq count data from `data/all_counts.txt`
2. Parse sample metadata
3. Perform QC and exploratory diagnostics on all samples
4. Evaluate pre-identified outlier samples through sensitivity analysis
5. Run filtered DESeq2 differential expression analysis
6. Export QC figures, DEG tables, and summary outputs to `results/`
7. Define high-confidence DEGs using adjusted p-value ≤ 0.005 and absolute log2 fold change > 2 .

Load packages

Project paths

Import count matrix

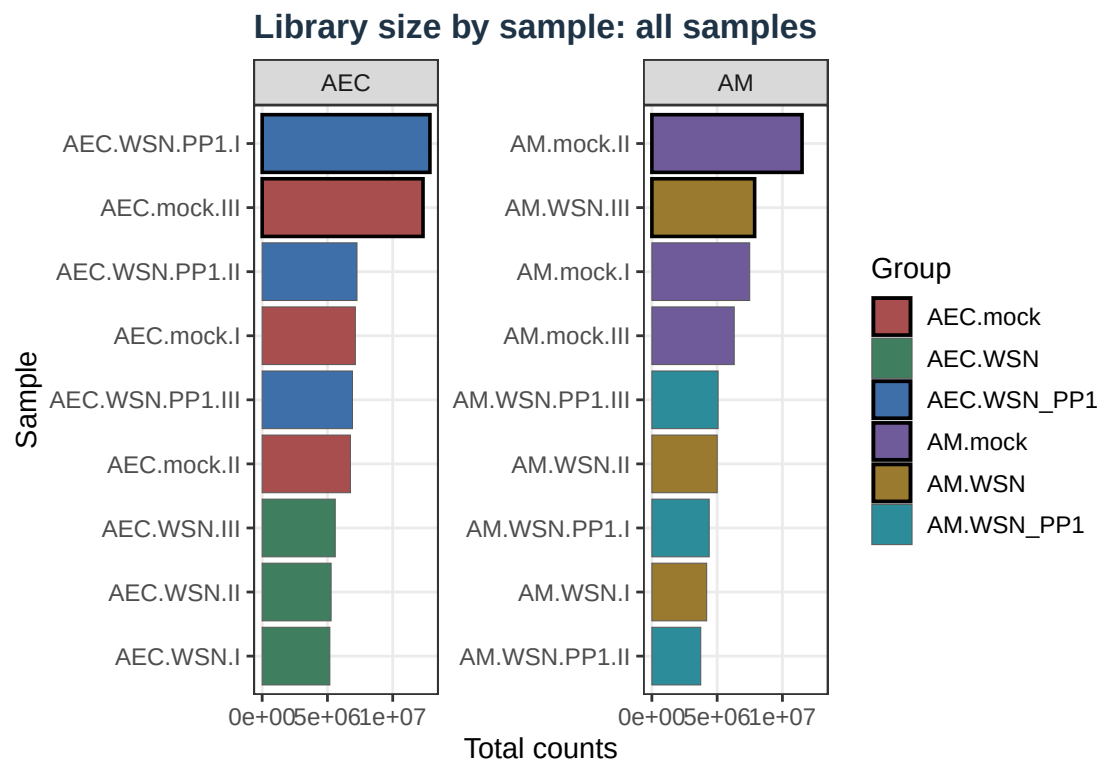
Build sample metadata

Optional gene annotation

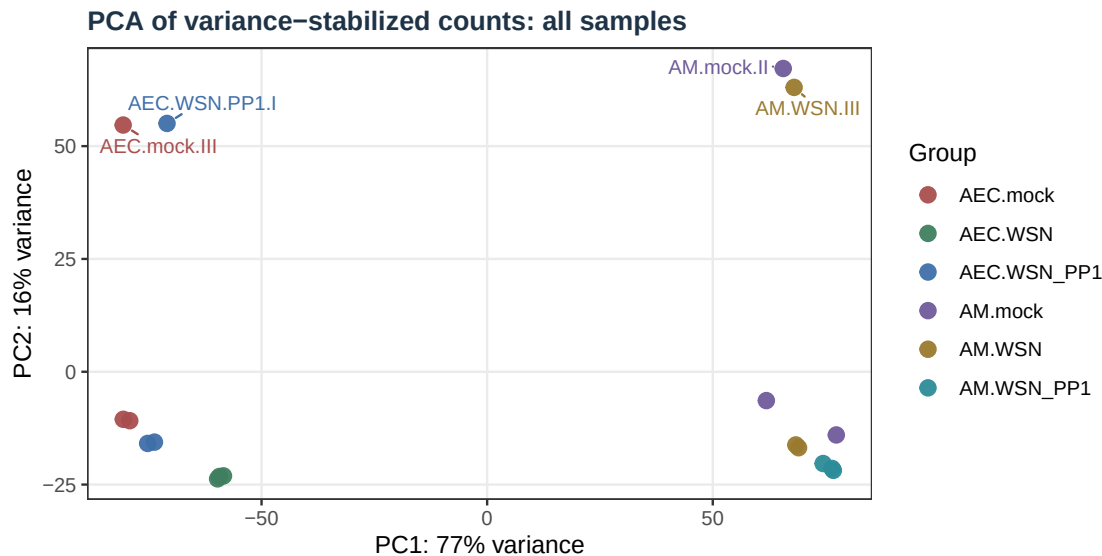
QC on all samples

Quality control is performed before excluding any samples. The outlier samples are flagged visually and retained in this section.

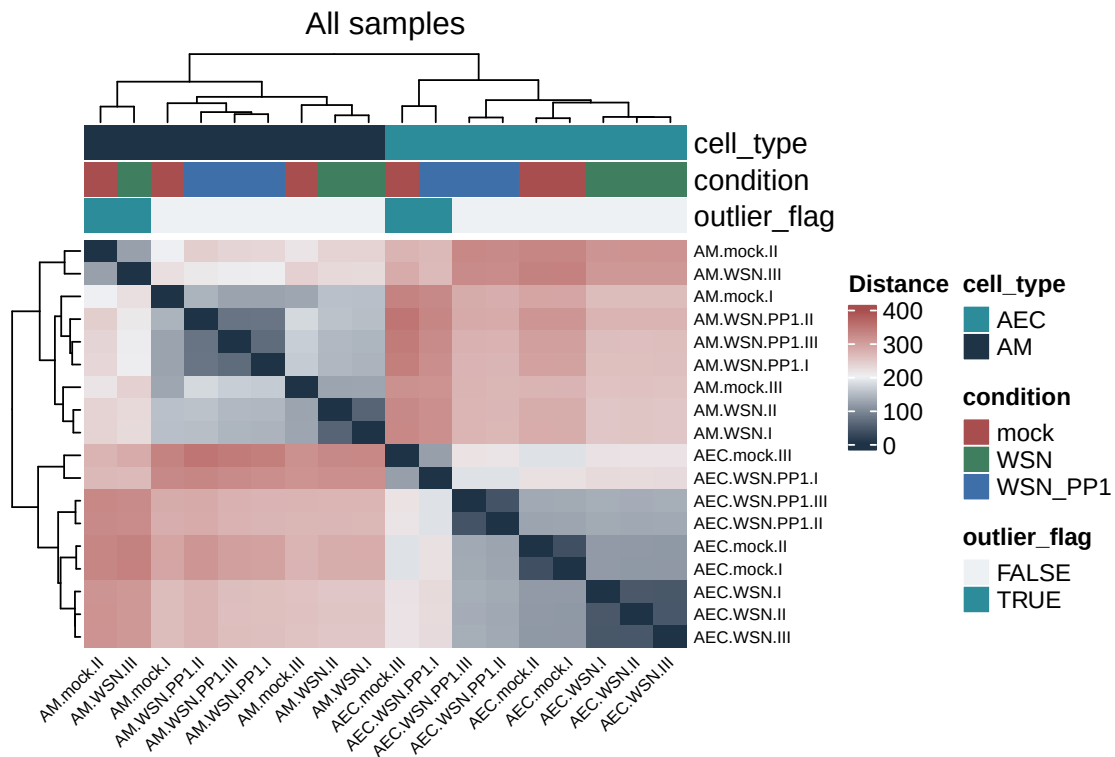
Library size



PCA: all samples



Sample distance heatmap: all samples



Helper functions

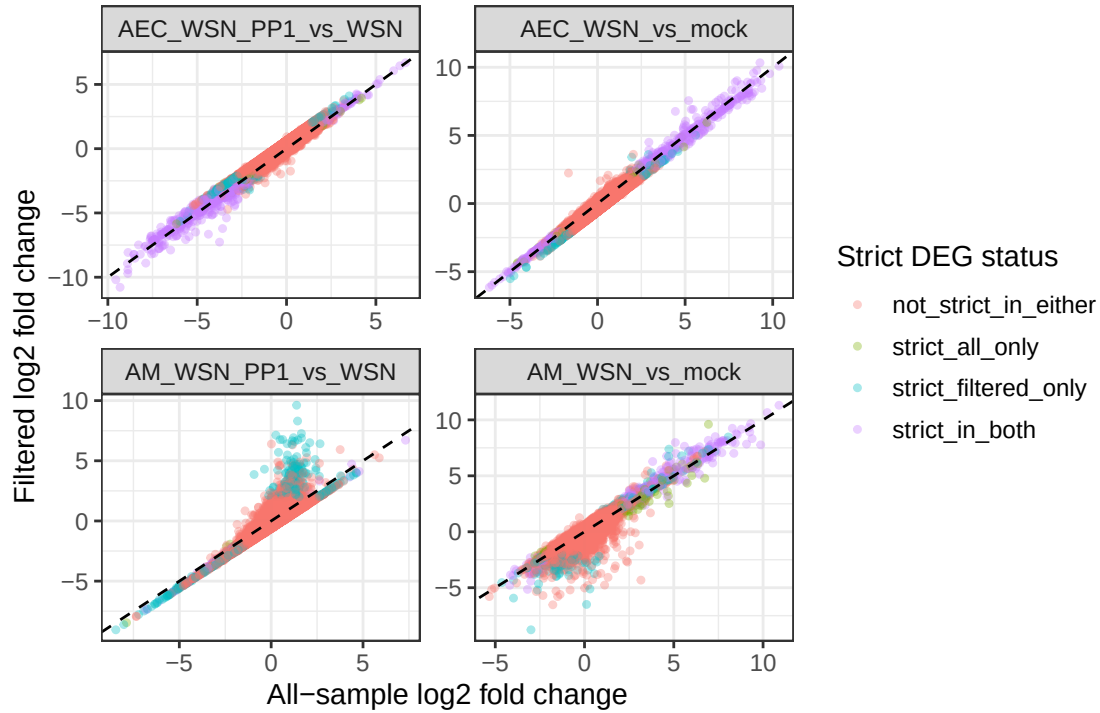
Sensitivity analysis

All-sample vs filtered DESeq2 results

This section evaluates whether the pre-specified outlier handling decision alters differential expression results. Both all-sample and filtered analyses are retained and exported for transparency.

High-confidence DEGs were defined using an adjusted p-value threshold ≤ 0.005 and an absolute log2 fold-change > 2 . These thresholds were used for DEG summaries, sensitivity comparisons, and downstream reporting exports.

Sensitivity analysis: all-sample vs filtered log2 fold change

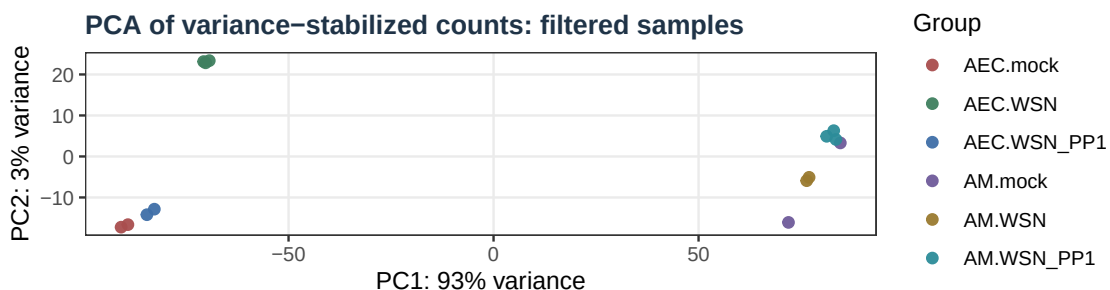


The sensitivity analysis shows that most high-confidence DEG calls are stable across filtering decisions, while selected contrasts, especially AM inhibitor-response comparisons, show larger shifts after outlier removal. This supports retaining both all-sample and filtered outputs for transparency.

Filtered DESeq2 workflow

The remaining workflow uses the filtered dataset, excluding the explicitly flagged outlier samples.

Filtered PCA

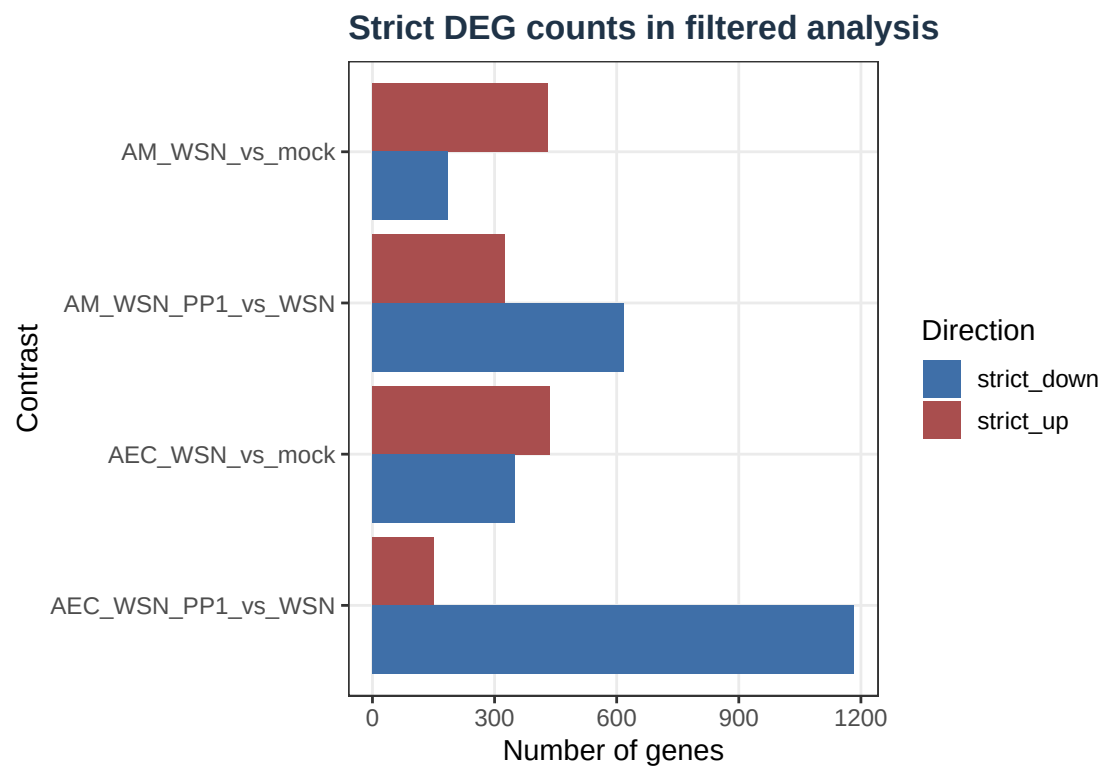


PCA of the filtered dataset showed strong separation by cell type along PC1 (93% of variance), while condition-related differences were captured along PC2 (3% of variance). These patterns support analyzing AEC and AM samples as separate pairwise condition comparisons.

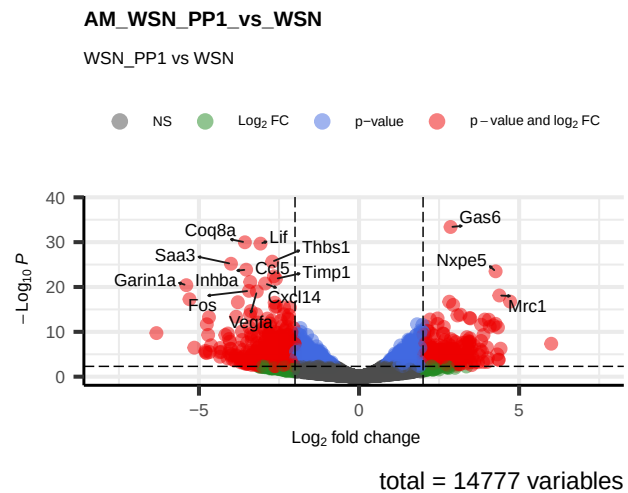
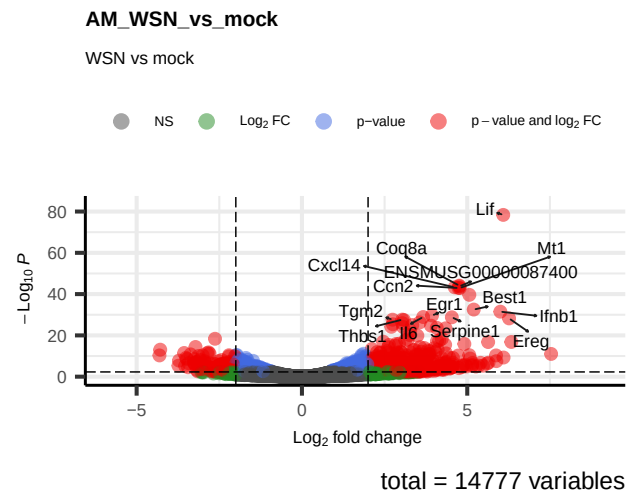
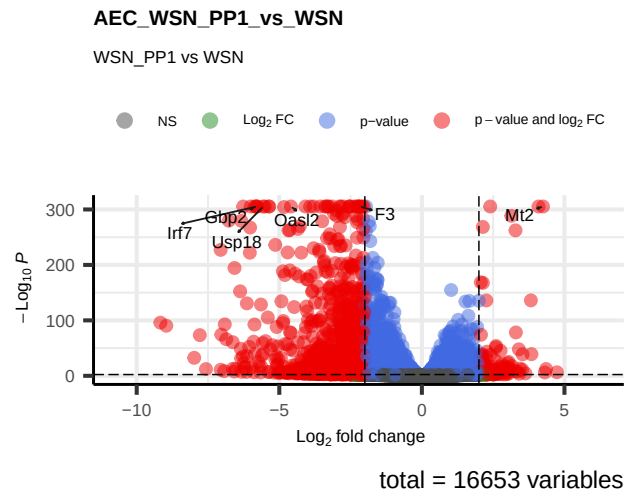
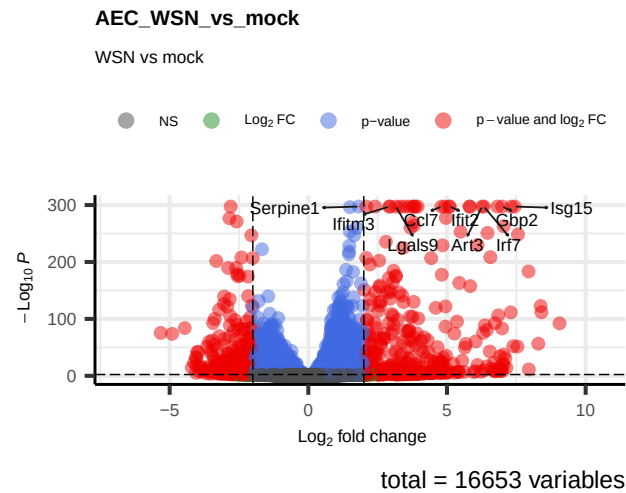
Export filtered DEG results

Filtered DEG visualizations

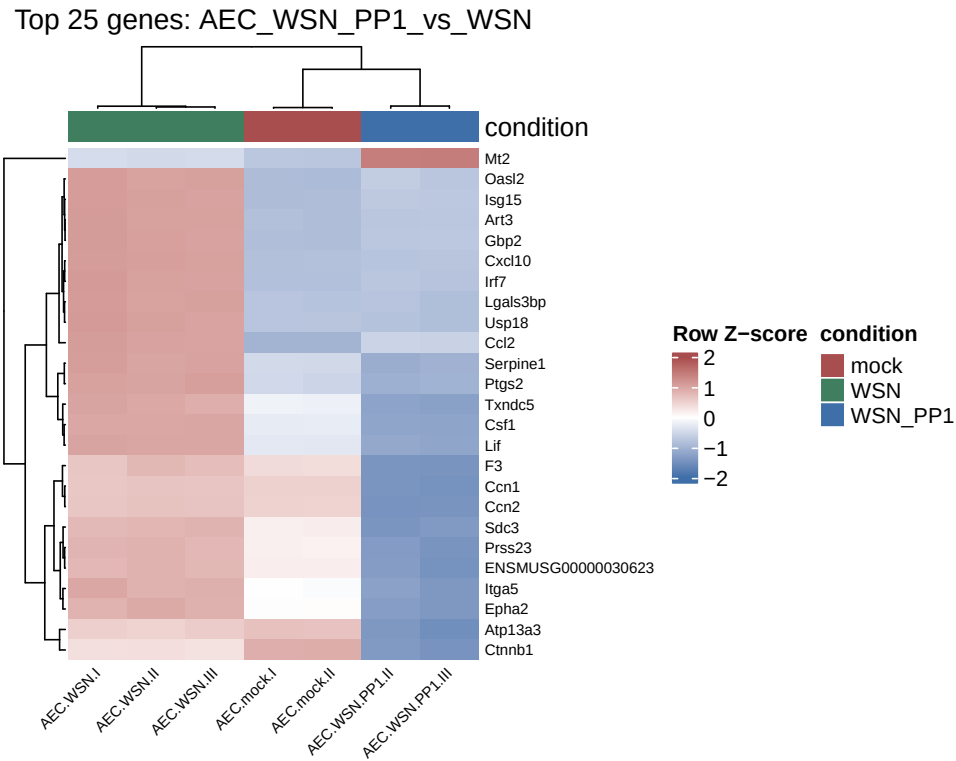
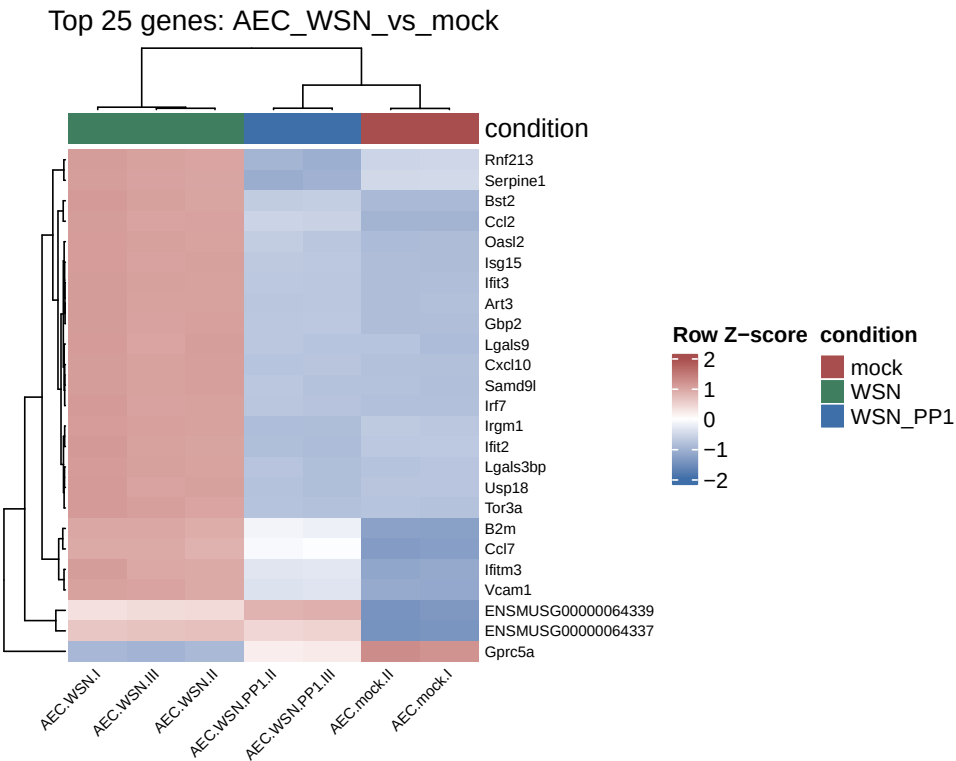
DEG count summary



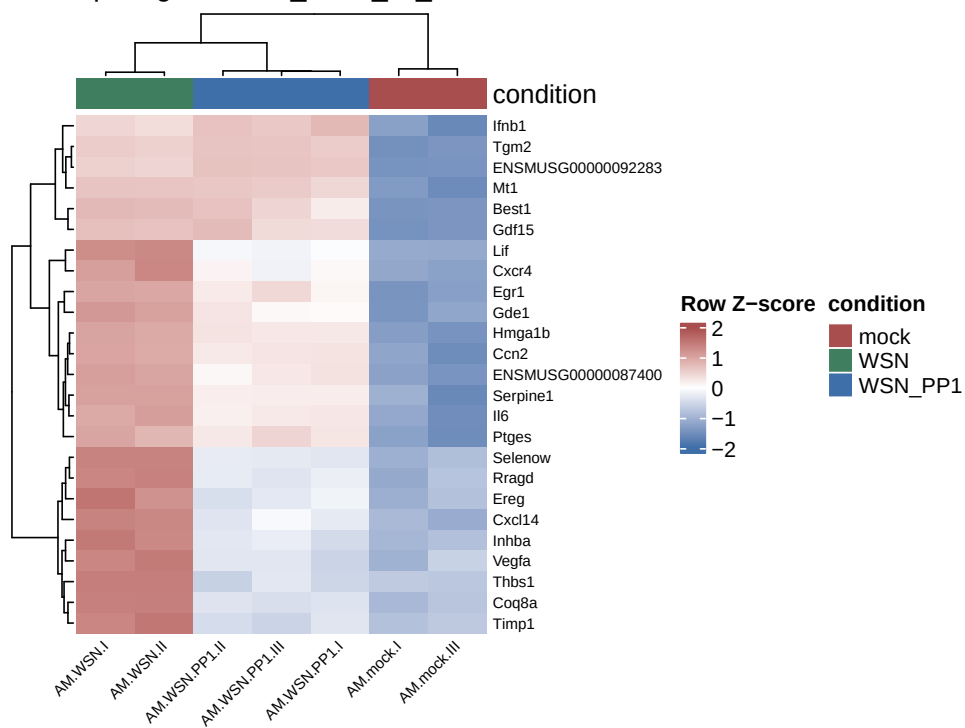
EnhancedVolcano plots with gene labels



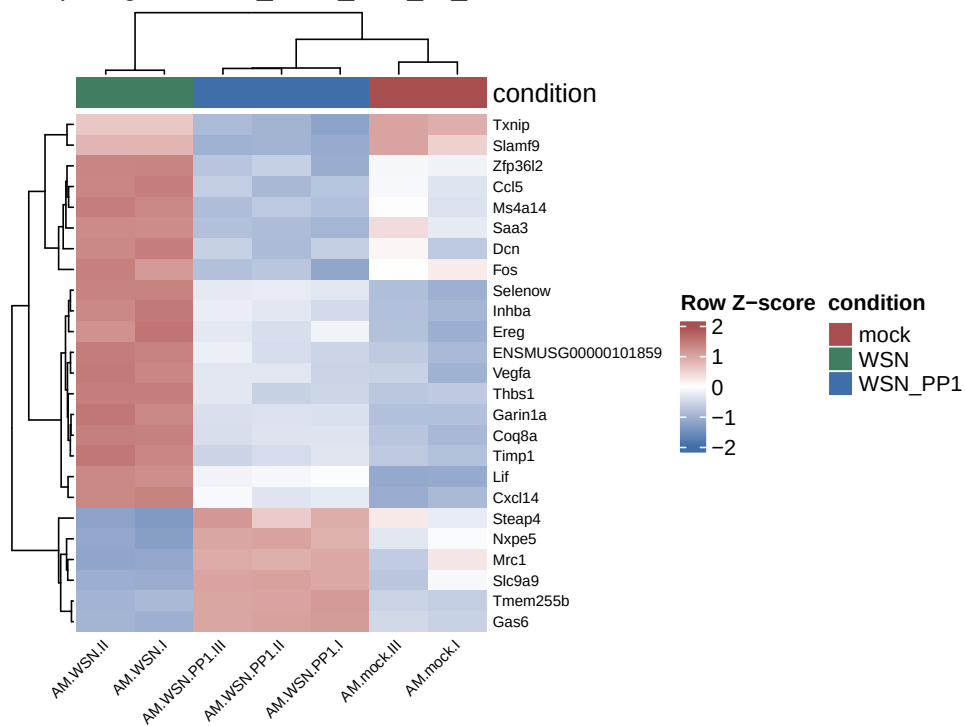
Heatmaps of top DEGs



Top 25 genes: AM_WSN_vs_mock



Top 25 genes: AM_WSN_PP1_vs_WSN



Conclusions

This workflow demonstrates a reproducible RNA-seq analysis and reporting pipeline integrating:

- QC and exploratory diagnostics
- transparent outlier sensitivity analysis
- DESeq2 differential expression workflows
- structured export of annotated DEG outputs
- downstream-ready reporting artifacts for Python-based aggregation and visualization

The analysis emphasizes reproducibility and transparent analytical decision-making rather than biological overinterpretation.

Reproducibility information

```
sessionInfo()
```

```
## R version 4.4.0 (2024-04-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/Los_Angeles
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats4    stats      graphics  grDevices  utils      datasets
## [8] methods  base
##
## other attached packages:
## [1] purrr_1.2.1          tibble_3.3.1
## [3] stringr_1.6.0        readr_2.2.0
## [5] tidyr_1.3.2          dplyr_1.2.0
## [7] pheatmap_1.0.13      org.Mm.eg.db_3.20.0
## [9] AnnotationDbi_1.68.0 circlize_0.4.18
## [11] ComplexHeatmap_2.22.0 EnhancedVolcano_1.24.0
## [13] ggrepel_0.9.8        ggplot2_4.0.2
## [15] DESeq2_1.46.0        SummarizedExperiment_1.36.0
## [17] Biobase_2.66.0       MatrixGenerics_1.18.1
## [19] matrixStats_1.5.0    GenomicRanges_1.58.0
## [21] GenomeInfoDb_1.42.3  IRanges_2.40.1
## [23] S4Vectors_0.44.0     BiocGenerics_0.52.0
##
```

```

## loaded via a namespace (and not attached):
## [1] DBI_1.3.0                rlang_1.1.7                magrittr_2.0.4
## [4] clue_0.3-68              GetoptLong_1.1.1           otel_0.2.0
## [7] compiler_4.4.0           RSQLite_2.4.6              png_0.1-9
## [10] systemfonts_1.3.2        vctrs_0.7.2                pkgconfig_2.0.3
## [13] shape_1.4.6.1            crayon_1.5.3               fastmap_1.2.0
## [16] XVector_0.46.0           labeling_0.4.3             utf8_1.2.6
## [19] rmarkdown_2.31          tzdb_0.5.0                 UCSC.utils_1.2.0
## [22] ragg_1.5.2               bit_4.6.0                  xfun_0.57
## [25] zlibbioc_1.52.0          cachem_1.1.0               jsonlite_2.0.0
## [28] blob_1.3.0               DelayedArray_0.32.0        BiocParallel_1.40.0
## [31] parallel_4.4.0           cluster_2.1.6              R6_2.6.1
## [34] stringi_1.8.7            RColorBrewer_1.1-3        Rcpp_1.1.1
## [37] iterators_1.0.14         knitr_1.51                 Matrix_1.7-0
## [40] tidyselect_1.2.1         rstudioapi_0.18.0         abind_1.4-8
## [43] yaml_2.3.12              doParallel_1.0.17         codetools_0.2-20
## [46] lattice_0.22-6           withr_3.0.2                KEGGREST_1.46.0
## [49] S7_0.2.1                 evaluate_1.0.5             Biostrings_2.74.1
## [52] pillar_1.11.1            foreach_1.5.2              generics_0.1.4
## [55] vroom_1.7.0              hms_1.1.4                  scales_1.4.0
## [58] glue_1.8.0               tools_4.4.0                locfit_1.5-9.12
## [61] colorspace_2.1-2         GenomeInfoDbData_1.2.13   cli_3.6.2
## [64] textshaping_1.0.5        S4Arrays_1.6.0            gtable_0.3.6
## [67] digest_0.6.39            SparseArray_1.6.2         rjson_0.2.23
## [70] farver_2.1.2             memoise_2.0.1             htmltools_0.5.9
## [73] lifecycle_1.0.5         httr_1.4.8                 GlobalOptions_0.1.4
## [76] bit64_4.6.0-1

```